

MUSTAFA YALVAÇ

Electrical & Electronics Engineer — E-Drive Development & EV Quality

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PROFESSIONAL PROFILE

Electrical and electronics engineer with over five years in electric powertrain development and automotive quality, promoted from motor design engineer to R&D Lead within four years. Designed, validated and industrialised electric machines for e-mobility and industrial programmes including TOGG, ISUZU, ASELSAN–KARSAN and Nordex. Now responsible at Hyundai Motor Manufacturing Turkey for charging-efficiency validation, ECU and CAN diagnostics, and quality assurance on new IONIQ models. Comfortable across the whole development chain — electromagnetic simulation, bench testing, industrialisation — and experienced in coordinating R&D, production, suppliers and certification bodies.

CORE COMPETENCIES

Electric Machine Design	IPM, synchronous reluctance & induction machines · electromagnetic topology · FEA-based optimisation
E-Mobility Systems	Electric Drive Unit (EDU) integration · traction motor sizing · inverter & battery supplier coordination
Test & Validation	Prototype validation · dyno & performance testing · charging efficiency · power measurement
Vehicle Diagnostics	CAN bus communication analysis · ECU fault diagnosis · Vector CANoe / CANalyzer
Quality & Compliance	Quality management systems · customer & certification audits · countermeasure & improvement processes
Leadership	R&D team building · project leadership · cross-functional coordination · training & technical documentation

PROFESSIONAL EXPERIENCE

Quality Engineer — EV Performance & Diagnostics	2025 – Present
Hyundai Motor Manufacturing Turkey · Kocaeli, Türkiye	
- Lead charging-efficiency and performance validation for the IONIQ series against global EV standards and internal quality gates. - Trace vehicle-level electrical faults to root cause with Vector CANoe and CANalyzer, analysing CAN traffic and ECU behaviour to close issues raised in production and validation. - Quantify charging performance, efficiency losses and power quality through high-precision measurement with Hioki power analysers. - Run quality assurance for the BJ new-model programme as project leader, aligning R&D and production teams from technical validation to launch readiness. - Manage the Global Quality Management System (GQMS) improvement pipeline, turning shop-floor findings into countermeasures that lift production efficiency and product quality.	
AEMOT Electric Motors & Generators	2021 – 2025
R&D Lead	2025
- Built and led a multidisciplinary R&D team spanning design, simulation, test and technical documentation, and set development priorities across the electric machine portfolio. - Owned the full development cycle for motors and generators, from electromagnetic design and simulation through prototyping, validation testing and industrialisation. - Converted a diesel forklift to electric drive: specified and developed the dedicated traction motor, then coordinated the inverter and battery suppliers through to a validated prototype.	

AEMOT Electric Motors & Generators (continued)

Senior R&D Engineer

2023 – 2025

- Led the electromagnetic design and optimisation of industrial and e-mobility machines, from concept to design freeze.
- Prepared the technical documentation for international certification and customer audits, and led the on-site reviews.
- Built expertise in electric drive unit (EDU) architecture for electric vehicles while remaining responsible for motor design, testing and commissioning.
- Delivered training on electric machine design and test methodology to universities, industry partners and internal engineering teams.

R&D Engineer

2021 – 2023

- Designed the electromagnetic topology of synchronous reluctance, induction and IPM machines in Motor-CAD, Altair Flux and Ansys Maxwell.
- Wrote the product datasheets and test reports for the electric machine range.
- Supported production and testing on the shop floor, including coil winding, rotor–stator assembly and prototype validation.

Electrical & Electronics Engineering Intern

Jun – Aug 2017

Koçak Engineering · Aksaray, Türkiye

- Assisted with electrical project drafting and on-site implementation, including control panel setup and internal wiring.

KEY PROJECTS

TOGG EV Traction Motor	Electrical work package owner
ASELSAN–KARSAN Motor Project	Electrical work package owner
ISUZU L7-Class Traction Motor	Project leader
Pilotcar Motor Project	Project leader
Nordex Wind Turbine Pitch & Yaw Motor	Electrical work package owner
Hyundai IONIQ 3 — EV Efficiency & Software Validation	Quality & diagnostics lead
Forklift Diesel-to-Electric Powertrain Conversion	Project leader — motor development & supplier coordination

TECHNICAL SKILLS

Electric Machine Design	Ansys Maxwell, Altair Flux, Motor-CAD, Simcenter SPEED
Diagnostics & Measurement	Vector CANoe, Vector CANalyzer, Hioki Power Analyzer
CAD	SolidWorks, AutoCAD
Systems & Reporting	IFS ERP, Microsoft Office

EDUCATION

B.Sc. Electrical and Electronics Engineering — Aksaray University	2015 – 2019
English Preparatory Year — Niğde University	2014 – 2015

LANGUAGES

Turkish	Native
English	Professional working proficiency